

Лабораторная работа №1

Выполнила: Брягиня Владислав 453503. Вариант 6.

Тема: Система контроля версий Git.

Цель: Познакомиться с возможностями и получить практические навыки работы с системой контроля версий Git.

Задание:

1. Подготовка рабочего окружения

a. Установка Git

```
[sunieq@MacBook-Air-Vladislav ~ % git --version  
git version 2.39.5 (Apple Git-154)
```

b. Выполнение базовых настроек

```
[sunieq@MacBook-Air-Vladislav ~ % git config --global user.name "Vladislav"  
[sunieq@MacBook-Air-Vladislav ~ % git config --global user.email "lounerexe13@gmail.com"]
```

2. Создание структуры папок (использовать командную строку) и поместить в репозиторий Git на своем локальном компьютере

Создание структуры папок:

```
[sunieq@MacBook-Air-Vladislav ~ % mkdir 453503_BRYAGINYA_6  
[sunieq@MacBook-Air-Vladislav ~ % cd 453503_BRYAGINYA_6  
[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % mkdir IGI  
[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % mkdir STRWEB  
[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % cd IGI
```

```
sunieq@MacBook-Air-Vladislav IGI % touch IGITest.txt  
sunieq@MacBook-Air-Vladislav IGI % mkdir LR1  
sunieq@MacBook-Air-Vladislav IGI % mkdir LR2  
sunieq@MacBook-Air-Vladislav IGI % mkdir LR3  
sunieq@MacBook-Air-Vladislav IGI % mkdir LR4  
sunieq@MacBook-Air-Vladislav IGI % mkdir LR5  
sunieq@MacBook-Air-Vladislav IGI % cd ..  
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % cd STRWEB  
sunieq@MacBook-Air-Vladislav STRWEB % touch STRWEBTest.txt  
sunieq@MacBook-Air-Vladislav STRWEB % mkdir LR1  
sunieq@MacBook-Air-Vladislav STRWEB % mkdir LR2  
sunieq@MacBook-Air-Vladislav STRWEB % mkdir LR3  
sunieq@MacBook-Air-Vladislav STRWEB % mkdir LR4
```

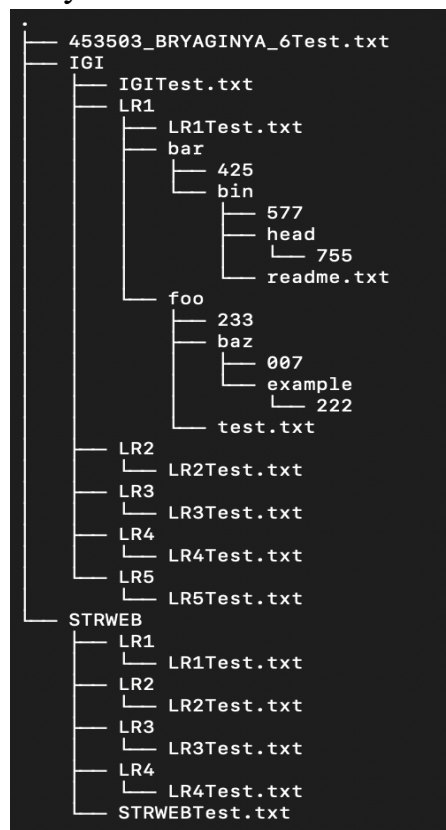
```
[sunieq@MacBook-Air-Vladislav LR1 % mkdir bar  
[sunieq@MacBook-Air-Vladislav LR1 % cd bar  
[sunieq@MacBook-Air-Vladislav bar % mkdir 425
```

```

sunieq@MacBook-Air-Vladislav bar % mkdir bin
sunieq@MacBook-Air-Vladislav bar % cd bin
sunieq@MacBook-Air-Vladislav bin % mkdir 577
sunieq@MacBook-Air-Vladislav bin % mkdir head
sunieq@MacBook-Air-Vladislav bin % cd head
sunieq@MacBook-Air-Vladislav head % mkdir 755
sunieq@MacBook-Air-Vladislav head % cd ..
sunieq@MacBook-Air-Vladislav bin % touch readme.txt
sunieq@MacBook-Air-Vladislav bin % echo "444" > readme.txt
sunieq@MacBook-Air-Vladislav bin % cd ..
sunieq@MacBook-Air-Vladislav bar % cd ..
sunieq@MacBook-Air-Vladislav LR1 % mkdir foo
sunieq@MacBook-Air-Vladislav LR1 % mkdir 233
sunieq@MacBook-Air-Vladislav LR1 % touch test.txt
sunieq@MacBook-Air-Vladislav LR1 % echo "707" > test.txt
sunieq@MacBook-Air-Vladislav LR1 % mkdir foo
sunieq@MacBook-Air-Vladislav LR1 % cd foo
sunieq@MacBook-Air-Vladislav foo % mkdir 233
sunieq@MacBook-Air-Vladislav foo % cd ..
sunieq@MacBook-Air-Vladislav LR1 % cd foo
sunieq@MacBook-Air-Vladislav foo % touch test.txt
sunieq@MacBook-Air-Vladislav foo % echo "707" > test.txt
sunieq@MacBook-Air-Vladislav foo % mkdir baz
sunieq@MacBook-Air-Vladislav foo % cd baz
sunieq@MacBook-Air-Vladislav baz % mkdir 007
sunieq@MacBook-Air-Vladislav baz % mkdir example
sunieq@MacBook-Air-Vladislav baz % cd example
sunieq@MacBook-Air-Vladislav example % mkdir 222

```

Результат:



Создание репозитория:

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git init
Initialized empty Git repository in /Users/sunieq/453503_BRYAGINYA_6/.git/
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .DS_Store
        453503_BRYAGINYA_6Test.txt
        IGI/
        STRWEB/

nothing added to commit but untracked files present (use "git add" to track)
```

3. Создайте репозиторий внутри папки LRI с помощью клонирования проекта с сайта [github.com](https://github.com/smartigaorg/geometric_lib)

(https://github.com/smartigaorg/geometric_lib)

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git clone https://github.com/smartigaorg/geometric_lib
Cloning into 'geometric_lib'...
remote: Enumerating objects: 35, done.
remote: Total 35 (delta 0), reused 0 (delta 0), pack-reused 35 (from 1)
Receiving objects: 100% (35/35), 4.69 KiB | 2.35 MiB/s, done.
Resolving deltas: 100% (5/5), done.
```

4. Выполните следующие действия, каждый раз проверяя состояние файлов в локальном репозитории до и после выполнения операции:

- a. сделать индексацию нескольких файлов в вашем Git-репозитории (git add ...)

```
sunieq@MacBook-Air-Vladislav foo % git add test.txt
[sunieq@MacBook-Air-Vladislav bin % git add readme.txt
sunieq@MacBook-Air-Vladislav bin % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   readme.txt
        new file:   ../../foo/test.txt
```

- b. внесите изменения в файлы

```
[sunieq@MacBook-Air-Vladislav foo % cat test.txt
707
[sunieq@MacBook-Air-Vladislav foo % echo "123" > test.txt
sunieq@MacBook-Air-Vladislav bin % cat readme.txt
222
sunieq@MacBook-Air-Vladislav bin % echo "234" > readme.txt
```

```

sunieq@MacBook-Air-Vladislav bin % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   readme.txt
    new file:   ../../foo/test.txt

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   readme.txt
    modified:   ../../foo/test.txt

```

- c. ВЫПОЛНИТЕ КОММИТЫ К ПРОИНДЕКСИРОВАННЫМ файлам

```

[sunieq@MacBook-Air-Vladislav bin % git commit -m "editing two files"
[main (root-commit) 59ae4d8] editing two files
 2 files changed, 2 insertions(+)
 create mode 100644 IGI/LR1/bar/bin/readme.txt
 create mode 100644 IGI/LR1/foo/test.txt

```

- d. добавьте и удалите файлы (один и группу файлов с определенным расширением)

> bar	Вчера, 22:02	-- Папка
> foo	Вчера, 22:06	-- Папка
image1.jpg	17 июля 2025 г., 05:11	28 КБ JPEG
image2.jpg	17 июля 2025 г., 05:10	68 КБ JPEG
image3.jpg	17 июля 2025 г., 05:10	58 КБ JPEG
LR1Test.txt	Вчера, 21:49	0 КБ Простой текст

```

[sunieq@MacBook-Air-Vladislav LR1 % git add '*.jpg'
[sunieq@MacBook-Air-Vladislav LR1 % git status
On branch main

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   image1.jpg
    new file:   image2.jpg
    new file:   image3.jpg

```

```

[sunieq@MacBook-Air-Vladislav LR1 % git rm image1.jpg -f
rm 'IGI/LR1/image1.jpg'

```

```

[sunieq@MacBook-Air-Vladislav LR1 % git rm '*.jpg' -f
rm 'IGI/LR1/image2.jpg'
rm 'IGI/LR1/image3.jpg'

```

```

sunieq@MacBook-Air-Vladislav LR1 % git status
On branch main

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   bar/bin/readme.txt
    modified:   foo/test.txt

```


- е. переименуйте файлы

```
sunieq@MacBook-Air-Vladislav LR1 % touch error
sunieq@MacBook-Air-Vladislav LR1 % git add error
sunieq@MacBook-Air-Vladislav LR1 % git mv error error
sunieq@MacBook-Air-Vladislav LR1 % git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   error
```

- ф. создайте для IGI и STRWEB, а также для каждой ЛР, вложенной в них, отдельные ветки, переместите папки и файлы на соответствующие ветки

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git switch -c IGI
Switched to a new branch 'IGI'
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git rm -r --cached -- STRWEB geometric_lib
rm 'STRWEB/LR1/LR1Test.txt'
rm 'STRWEB/LR2/LR2Test.txt'
rm 'STRWEB/LR3/LR3Test.txt'
rm 'STRWEB/LR4/LR4Test.txt'
rm 'STRWEB/STRWEBTest.txt'
rm 'geometric_lib'
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git commit -m "keep only IGI"
[IGI 20c509f] keep only IGI
6 files changed, 1 deletion(-)
delete mode 100644 STRWEB/LR1/LR1Test.txt
delete mode 100644 STRWEB/LR2/LR2Test.txt
delete mode 100644 STRWEB/LR3/LR3Test.txt
delete mode 100644 STRWEB/LR4/LR4Test.txt
delete mode 100644 STRWEB/STRWEBTest.txt
delete mode 160000 geometric_lib
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git status
On branch IGI
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        STRWEB/
        geometric_lib/
```

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git switch -c IGI_LR1
Switched to a new branch 'IGI_LR1'
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git rm -r --cached -- IGI/LR2 IGI/LR3 IGI/LR4 IGI/LR5
rm 'IGI/LR2/LR2Test.txt'
rm 'IGI/LR3/LR3Test.txt'
rm 'IGI/LR4/LR4Test.txt'
rm 'IGI/LR5/LR5Test.txt'
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git commit -m "keep only IGI/LR1"
```

```
Switched to a new branch 'STRWEB'
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git rm -r --cached -- IGI geometric_lib
rm 'IGI/.DS_Store'
rm 'IGI/IGITest.txt'
rm 'IGI/LR1/.DS_Store'
rm 'IGI/LR1/LR1Test.txt'
rm 'IGI/LR1/bar/bin/readme.txt'
rm 'IGI/LR1/error'
rm 'IGI/LR1/foo/test.txt'
rm 'IGI/LR2/LR2Test.txt'
rm 'IGI/LR3/LR3Test.txt'
rm 'IGI/LR4/LR4Test.txt'
rm 'IGI/LR5/LR5Test.txt'
rm 'geometric_lib'
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git commit -m "STRWEB: keep only STRWEB"
[STRWEB 885f037] STRWEB: keep only STRWEB
12 files changed, 3 deletions(-)
delete mode 100644 IGI/.DS_Store
delete mode 100644 IGI/IGITest.txt
delete mode 100644 IGI/LR1/.DS_Store
delete mode 100644 IGI/LR1/LR1Test.txt
delete mode 100644 IGI/LR1/bar/bin/readme.txt
delete mode 100644 IGI/LR1/error
delete mode 100644 IGI/LR1/foo/test.txt
delete mode 100644 IGI/LR2/LR2Test.txt
delete mode 100644 IGI/LR3/LR3Test.txt
delete mode 100644 IGI/LR4/LR4Test.txt
delete mode 100644 IGI/LR5/LR5Test.txt
delete mode 160000 geometric_lib
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git ls-files
```

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git switch STRWEB
git switch -c STRWEB_LR1
git rm -r --cached -- STRWEB/LR2 STRWEB/LR3 STRWEB/LR4
git commit -m "STRWEB_LR1: keep only STRWEB/LR1"
```

```
Switched to a new branch 'STRWEB_LR1'
rm 'STRWEB/LR2/LR2Test.txt'
rm 'STRWEB/LR3/LR3Test.txt'
rm 'STRWEB/LR4/LR4Test.txt'
[STRWEB_LR1 e3d91e8] STRWEB_LR1: keep only STRWEB/LR1
3 files changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 STRWEB/LR2/LR2Test.txt
delete mode 100644 STRWEB/LR3/LR3Test.txt
delete mode 100644 STRWEB/LR4/LR4Test.txt
```

- g. ВЫВЕДИТЕ ИСТОРИЮ КОММИТОВ: ПОЛНОСТЬЮ И С ОГРАНИЧЕНИЕМ ВЫВОДА – ОТОБРАЗИТЬ ТОЛЬКО ПОСЛЕДНИЕ, ДО И ПОСЛЕ УКАЗАННОЙ ДАТЫ, ПО АВТОРУ ФАЙЛОВ, КОММИТОВ (флаги n, since, after, until, before, author, committer, grep) <https://code.mu/ru/tool/git/basis/limit-output-commit-history-flags/>

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log
commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300
```

add all files

```
commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300
```

editing error to error

```
commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300
```

editing two files

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --oneline
```

```
2e1534a (HEAD -> main) add all files
e35a274 editing error to error
59ae4d8 editing two files
```

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log -n 2
```

```
commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300
```

add all files

```
commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300
```

editing error to error

```

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --since="2025-02-18"

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

```

```

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --until="2026-02-20"

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

```

```

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --author="Vladislav"

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

```

```

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --committer="Vladislav"

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

```

```

[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --grep="add"

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

```


- h. исследуйте флаги для git log: p, stat, shortstat, name-only, name-status, relative-date, pretty, graph.

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --stat

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

.DS_Store | Bin 0 -> 10244 bytes
453503_BRYAGINYA_6Test.txt | 0
IGI/.DS_Store | Bin 0 -> 10244 bytes
IGI/IGITest.txt | 0
IGI/LR1/.DS_Store | Bin 0 -> 6148 bytes
IGI/LR1/LR1Test.txt | 0
IGI/LR1/bar/bin/readme.txt | 2 +-
IGI/LR1/foo/test.txt | 2 +-
IGI/LR2/LR2Test.txt | 0
IGI/LR3/LR3Test.txt | 0
IGI/LR4/LR4Test.txt | 0
IGI/LR5/LR5Test.txt | 0
STRWEB/LR1/LR1Test.txt | 0
STRWEB/LR2/LR2Test.txt | 0
STRWEB/LR3/LR3Test.txt | 0
STRWEB/LR4/LR4Test.txt | 0
STRWEB/STRWEBTest.txt | 0
geometric_lib | 1 +
18 files changed, 3 insertions(+), 2 deletions(-)

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

IGI/LR1/error | 0
1 file changed, 0 insertions(+), 0 deletions(-)

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

IGI/LR1/bar/bin/readme.txt | 1 +
IGI/LR1/foo/test.txt | 1 +
2 files changed, 2 insertions(+)

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --shortstat

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

18 files changed, 3 insertions(+), 2 deletions(-)

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

1 file changed, 0 insertions(+), 0 deletions(-)

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

2 files changed, 2 insertions(+)
```



```

2 files changed, 2 insertions(+), 2 deletions(-)
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --shortstat

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

18 files changed, 3 insertions(+), 2 deletions(-)

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

1 file changed, 0 insertions(+), 0 deletions(-)

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

2 files changed, 2 insertions(+)

```

```

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --name-only

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

.DS_Store
453503_BRYAGINYA_6Test.txt
IGI/.DS_Store
IGI/IGITest.txt
IGI/LR1/.DS_Store
IGI/LR1/LR1Test.txt
IGI/LR1/bar/bin/readme.txt
IGI/LR1/foo/test.txt
IGI/LR2/LR2Test.txt
IGI/LR3/LR3Test.txt
IGI/LR4/LR4Test.txt
IGI/LR5/LR5Test.txt
STRWEB/LR1/LR1Test.txt
STRWEB/LR2/LR2Test.txt
STRWEB/LR3/LR3Test.txt
STRWEB/LR4/LR4Test.txt
STRWEB/STRWEBTest.txt
geometric_lib

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

IGI/LR1/error

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

```

```

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --relative-date

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: 2 hours ago

    add all files

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: 2 hours ago

    editing error to error

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: 3 hours ago

    editing two files

```

```

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --name-status

commit 2e1534afd643c0c4af07ad2e063d20cd87f85a84 (HEAD -> main)
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:52:12 2026 +0300

    add all files

A       .DS_Store
A       453503_BRYAGINYA_6Test.txt
A       IGI/.DS_Store
A       IGI/IGITest.txt
A       IGI/LR1/.DS_Store
A       IGI/LR1/LR1Test.txt
M       IGI/LR1/bar/bin/readme.txt
M       IGI/LR1/foo/test.txt
A       IGI/LR2/LR2Test.txt
A       IGI/LR3/LR3Test.txt
A       IGI/LR4/LR4Test.txt
A       IGI/LR5/LR5Test.txt
A       STRWEB/LR1/LR1Test.txt
A       STRWEB/LR2/LR2Test.txt
A       STRWEB/LR3/LR3Test.txt
A       STRWEB/LR4/LR4Test.txt
A       STRWEB/STRWEBTest.txt
A       geometric_lib

commit e35a2746b0cf9631319c4ca32011c7d3211ad645
Author: Vladislav <lounerexe13@gmail.com>
Date: Thu Feb 19 00:47:38 2026 +0300

    editing error to error

A       IGI/LR1/error

commit 59ae4d849ee63068a2849655c1a6fb8b9cc4ed4c
Author: Vladislav <lounerexe13@gmail.com>
Date: Wed Feb 18 23:32:47 2026 +0300

    editing two files

A       IGI/LR1/bar/bin/readme.txt
A       IGI/LR1/foo/test.txt

```

```

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --pretty=format:"%h - %an, %ar : %s"

2e1534a - Vladislav, 2 hours ago : add all files
e35a274 - Vladislav, 2 hours ago : editing error to error
59ae4d8 - Vladislav, 3 hours ago : editing two files

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --graph --oneline --all

* 0d68388 (STRWEB_LR4) STRWEB_LR4: keep only STRWEB/LR4
| * 3293221 (STRWEB_LR3) STRWEB_LR3: keep only STRWEB/LR3
| /
| * bf60c07 (STRWEB_LR2) STRWEB_LR2: keep only STRWEB/LR2
| /
| * e3d91e8 (STRWEB_LR1) STRWEB_LR1: keep only STRWEB/LR1
| /
* 885f037 (STRWEB) STRWEB: keep only STRWEB
| * ced6274 (IGI_LR5) IGI_LR5: keep only IGI/LR5
| | * 9dd04a7 (IGI_LR4) IGI_LR4: keep only IGI/LR4
| | /
| | * 6b3cdfd (IGI_LR3) IGI_LR3: keep only IGI/LR3
| | /
| | * b4cbbb4 (IGI_LR2) keep only IGI/LR2
| | /
| | * 150bcc7 (IGI_LR1) keep only IGI/LR1
| | /
| * 20c509f (IGI) keep only IGI
| /
* 2e1534a (HEAD -> main) add all files
* e35a274 editing error to error
* 59ae4d8 editing two files

```

- i. найдите разницу в изменениях для каждого коммита

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log -p --oneline

2e1534a (HEAD -> main) add all files
diff --git a/.DS_Store b/.DS_Store
new file mode 100644
index 0000000..255e8ef
Binary files /dev/null and b/.DS_Store differ
diff --git a/453503_BRYAGINYA_6Test.txt b/453503_BRYAGINYA_6Test.txt
new file mode 100644
index 0000000..e69de29
diff --git a/IGI/.DS_Store b/IGI/.DS_Store
new file mode 100644
index 0000000..38561ab
Binary files /dev/null and b/IGI/.DS_Store differ
diff --git a/IGI/IGITest.txt b/IGI/IGITest.txt
new file mode 100644
index 0000000..e69de29
diff --git a/IGI/LR1/.DS_Store b/IGI/LR1/.DS_Store
new file mode 100644
index 0000000..243181e
Binary files /dev/null and b/IGI/LR1/.DS_Store differ
diff --git a/IGI/LR1/LR1Test.txt b/IGI/LR1/LR1Test.txt
new file mode 100644
index 0000000..e69de29
diff --git a/IGI/LR1/bar/bin/readme.txt b/IGI/LR1/bar/bin/readme.txt
index 1e6fd03..7b5813c 100644
--- a/IGI/LR1/bar/bin/readme.txt
+++ b/IGI/LR1/bar/bin/readme.txt
@@ -1,1 @@
-444
+234
diff --git a/IGI/LR1/foo/test.txt b/IGI/LR1/foo/test.txt
index 8b8e485..190a180 100644
--- a/IGI/LR1/foo/test.txt
+++ b/IGI/LR1/foo/test.txt
@@ -1,1 @@
-707
+123
diff --git a/IGI/LR2/LR2Test.txt b/IGI/LR2/LR2Test.txt
new file mode 100644
index 0000000..e69de29
diff --git a/IGI/LR3/LR3Test.txt b/IGI/LR3/LR3Test.txt
new file mode 100644
index 0000000..e69de29
diff --git a/IGI/LR4/LR4Test.txt b/IGI/LR4/LR4Test.txt
new file mode 100644
index 0000000..e69de29
diff --git a/IGI/LR5/LR5Test.txt b/IGI/LR5/LR5Test.txt
new file mode 100644
index 0000000..e69de29
diff --git a/STRWEB/LR1/LR1Test.txt b/STRWEB/LR1/LR1Test.txt
new file mode 100644
index 0000000..e69de29
```

- j. создайте новый файл test.txt и вставьте в него строчку 'номер варианта'. затем сделайте коммит и отмените его

```
[sunieq@MacBook-Air-Vladislav LR1 % echo "6" > test.txt
[sunieq@MacBook-Air-Vladislav LR1 % git add test.txt
[sunieq@MacBook-Air-Vladislav LR1 % git commit -m "add text file with variant"
[IGI_LR1 dc2a63a] add text file with variant
1 file changed, 1 insertion(+)
create mode 100644 IGI/LR1/test.txt
[sunieq@MacBook-Air-Vladislav LR1 % git reset --soft HEAD~1
[sunieq@MacBook-Air-Vladislav LR1 % git status
On branch IGI_LR1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   test.txt
```

- k. проиндексируйте файл в вашем Git-репозитории, затем отмените его индексацию

```
[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % echo "124432" > test.txt
[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git add test.txt
[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git restore --staged test.txt
[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git status
On branch main
Changes to be committed:
```



```
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  test.txt
```

1. внесите изменения в файл в вашем Git-репозитории и затем верните его в исходное состояние.

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % echo "99999" > test.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   .DS_Store
    modified:   IGI/.DS_Store
    modified:   test.txt

no changes added to commit (use "git add" and/or "git commit -a")
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % cat test.txt
99999
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git restore test.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % cat test.txt
124432
```

5. Выполните следующие действия с удаленным репозиторием:

- a. создайте удаленный репозиторий (на github.com)
- b. просмотрите информацию о нем, внесите изменения и опять просмотрите (remote, fetch).

```
[sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git remote -v
origin  https://github.com/sunIQ22/IGI_LR1.git (fetch)
origin  https://github.com/sunIQ22/IGI_LR1.git (push)
```

- c. измените несколько файлов в вашем проекте и перенесите их в удаленный репозиторий

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % ls
453503_BRYAGINYA_6Test.txt  IGI  STRWEB  test.txt  test2.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % echo "update" > test.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % echo "update" > test2.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git add test.txt test2.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git commit -m "update two files"
[main 180b5b8] update two files
 2 files changed, 2 insertions(+), 2 deletions(-)
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 285 bytes | 285.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/sunIQ22/IGI_LR1.git
 864c1d7..180b5b8  main -> main
```

- d. переименуйте свой удаленный репозиторий

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git remote set-url origin https://github.com/sunIQ22/IGI_LR1_NEW_NAME
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git remote -v

origin  https://github.com/sunIQ22/IGI_LR1_NEW_NAME (fetch)
origin  https://github.com/sunIQ22/IGI_LR1_NEW_NAME (push)
```

- e. получите данные из удаленного репозитория

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git fetch
```

- f. просмотреть историю изменений

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git log --oneline
180b5b8 (HEAD -> main, origin/main) update two files
864c1d7 create test2.txt
ef7e521 Remove DS_Store from repo
5936e68 Add .gitignore for DS_Store
66b5888 Move geometric_lib to IGI/LR1
7b9ea2b add test.txt
2e1534a add all files
e35a274 editing error to error
59ae4d8 editing two files
```

- g. создайте дополнительную ветку 1 в удаленном репозитории, перенесите в нее два файла из другой ветки, удалите один из файлов

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git checkout IGI -- IGI/LR1/LR1Test.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git checkout IGI -- IGI/LR1/foo/test.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git rm IGI/LR1/foo/test.txt

rm 'IGI/LR1/foo/test.txt'
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git commit -m "Add two files from IGI and remove one"

[branch1 00d8c26] Add two files from IGI and remove one
1 file changed, 1 deletion(-)
delete mode 100644 IGI/LR1/foo/test.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git push -u origin branch1
```

- h. создайте дополнительную ветку 2 в удаленном репозитории, перенесите в нее два файла из вашего проекта, слить изменения в основную ветку, удалить дополнительную ветку 2

```
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % echo "branch2 change 1" >> test.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % echo "branch2 change 2" >> test2.txt

sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git add test.txt test2.txt
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git commit -m "branch2: update two files"
[branch2 ee2cf9a] branch2: update two files
2 files changed, 2 insertions(+)
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git push -u origin branch2

Enumerating objects: 7, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 363 bytes | 363.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'branch2' on GitHub by visiting:
remote:   https://github.com/suniek22/IGI_LR1_NEW_NAME/pull/new/branch2
remote:
To https://github.com/suniek22/IGI_LR1_NEW_NAME
 * [new branch]      branch2 -> branch2
branch 'branch2' set up to track 'origin/branch2'.
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git merge branch2
Updating 180b5b8..ee2cf9a
Fast-forward
 test.txt | 1 +
 test2.txt | 1 +
 2 files changed, 2 insertions(+)
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git push
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/suniek22/IGI_LR1_NEW_NAME
 180b5b8..ee2cf9a  main -> main
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git branch -d branch2
Deleted branch branch2 (was ee2cf9a).
sunieq@MacBook-Air-Vladislav 453503_BRYAGINYA_6 % git push origin --delete branch2
To https://github.com/suniek22/IGI_LR1_NEW_NAME
 - [deleted]      branch2
```

6. Сгенерируйте в DeepSeek файл html и файл css для сайта визитки компании или человека с областью занятости на выбор:

```
sunieq@MacBook-Air-Vladislav business-card-site % nano index.html
sunieq@MacBook-Air-Vladislav business-card-site % nano styles.css
sunieq@MacBook-Air-Vladislav business-card-site % git init
git add index.html styles.css
git commit -m "Add landing page"

Initialized empty Git repository in /Users/sunieq/business-card-site/.git/
[main (root-commit) ddb14cc] Add landing page
2 files changed, 989 insertions(+)
 create mode 100644 index.html
 create mode 100644 styles.css
sunieq@MacBook-Air-Vladislav business-card-site % git branch -M main
git remote add origin https://github.com/suniek22/business-card-site.git
git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 7.53 KiB | 7.53 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/suniek22/business-card-site.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

